Urban Tree Canopy (UTC) and Total Vegetation Estimate Workflow

UTC estimates

- 11.9 % canopy (USDA Forest Service 2007)
- 4,148 acres or 13.7% canopy (SF Planning Dept 2013, based on 2010 data)
- 5,276 acres or 17.5% canopy (this estimate based on 2022/2023 data)

Total Vegetation estimate

- 7,602 acres (NAIP May 2022)
- 9,575 acres (Sentinel-2, May 2022)

Link to files: **UTC and Total Vegetation Files**

Link to GEE repo:
https://code.earthengine.google.com/?accept_repo=users/harpecas/ReimaginingSanFrancisco

2022 NAIP - National Agricultural Imagery Program

- Through the USDA, aerial imagery collected via aircraft during growing season
- Orthorectified raster data
- Flown between May 18-19 2022
- 0.6 to 1 meter resolution
- 4 spectral bands: Blue, Green, Red, and Near Infrared (NIR)
- Temporal resolution for SF has been every 2 years since 2010. The last acquisition was in 2024, but that imagery is not yet available

LiDAR data

- Used in Urban Tree Canopy estimate
- Flown April 20 2023
- Average density of 50 pts per sq meter
- Maximum Surface Height Raster (MSH) - natural and built up features of 0.5m pixels
- From this I used a derived data product called a Height Above Ground raster resampled to 2 meters (Source: https://doi.org/10.5281/zenodo.15443630)

Sentinel-2 data

- 2 satellites that collectively fly over SF every 5 days
- 10 meter resolution
- Data available from 2017-present
- Imagery masked for building footprints and clouds

Study Area

- Using the same value for the total area of San Francisco from the 2013 estimate, **30,178 acres**
- City of San Francisco including Alcatraz, Treasure Island/Yerba Buena Island but not including the Farallon Islands.

NAIP Total Vegetation Workflow

This workflow can be used to estimate total vegetation for any year where NAIP imagery was collected for the state of California. The last acquisition was in 2024, but that imagery is not yet available in ArcGIS or Google Earth Engine.

Calculate total Normalized Difference Vegetation Index (NDVI) using the red and near-infrared bands and use that to estimate total area, in acres, of vegetation for the city.

NDVI = Normalized Difference Vegetation Index - widely used metric for quantifying vegetation. Chlorophyll strongly absorbs visible light in the 400-700 nanometer range for photosynthesis, and the cell structure of leaves strongly reflects near-infrared light in the range of 700-1100 nanometers. It is calculated as (NIR - Red) / (NIR + Red)

Healthy vegetation will have low values of reflectance in the visible light range the Red band) and high values of reflectance in the near-infrared range (the NIR band). As a result, dense healthy vegetation will have an NDVI value close to 1, dry non-vegetated land will be near 0.

The methodology described in the UTC analysis performed in 2013 uses a threshold of 0.2
An NDVI value of >0.2 is a standard threshold for distinguishing vegetation from other land types, i.e. buildings and roads. Anything greater than 0.2 = vegetation, anything equal to or less than 0.2 is classified as non vegetation.

Using an NDVI threshold of 0.2 on the 2022 data, I found areas of vegetation and tree canopy visible in the true color map imagery that were being missed.

Screenshots: left - true color image of Sunset Blvd, right - areas counting as vegetation masked in green using an NDVI of 0.2. Many trees, vegetated areas not being included

There could be multiple reasons why vegetation is not being counted with an NDVI value > 0.2. In an urban environment, shadows cast by tall buildings, the topography, or the tree canopy itself could be shading out and lowering reflectance values. Also, the resolution of the imagery collected changed between the data collected in 2010 and used in the 2013 assessment and the data collected in 2022.

2010 NAIP is 4 band 1 meter, 8 bit

2022 NAIP is 4 band 0.6 meter (60 cm), 8 bit resampled from 10-12 bit raw

I found an NDVI threshold of 0.05 works better for the 2022 data. It captures more vegetation, including areas of sparse vegetation without including false positives, like roads or paved surfaces.

Screenshots: left - true color image of Sunset Blvd, right - areas counting as vegetation masked in green using an NDVI of 0.05.

All pixels classified as vegetation are then summed and converted to an estimate in both square meters and acres.

Results in an estimate of 7,602 acres of total vegetation. Using the same value for the total area of San Francisco from the 2013 estimate, 30,178 acres, results in a percentage of 25.2% total vegetation.

UTC Workflow

UTC estimates combine NDVI data with height data to classify trees from other vegetation.

First, using the same vegetation workflow as described above, pixels in the image are classified as vegetation or non-vegetation using an NDVI threshold of 0.05.

Second categorize LiDAR data into the same 3 height classes used in 2013 (knowledge based classification system)

- Class 1: < 1' (Grass, pavement, soil, open water)
- Class 2: 1' – 8' (Transitional layer, shrubs, cars,)
- Class 3: > 8' (Trees, buildings)

The initial LiDAR data was transformed into a Height Above Ground raster. Each point is associated with a height and a layer created where all points are categorized into three height classes.

Finally, tree canopy is defined as an NDVI value greater than 0.05 and a height above ground value greater than 8 feet. Every point on the map meeting both of those criteria is classified as tree canopy.

All pixels classified as tree canopy are summed and converted to an estimate in both square meters and acres. The Urban Tree Canopy area in acres is estimated to be 5,276 acres. Using the same value for the total area of San Francisco from the 2013 estimate, 30,178 acres, results in a percentage of 17.5% UTC, with tree canopy accounting for 69% of all vegetated areas.

Sentinel-2 Total Vegetation Workflow

What the GEE code does:

- For specified start and end dates, it will create a median composite of NDVI (vegetation index)
    - Not all images will have good data due to cloud cover. Compositing over a month increases the chance that for any given pixel you'll have a representative value
- Provide an estimate of total vegetation that meets the provided threshold of NDVI in square meters and acres
- Provides a raster file that can be loaded into a GIS platform
    - New script calculates NDVI as an integer and resulting GeoTIFF has NDVI values that range from -0.2 to 807 (theoretical range is -1000 to 1000)
    - Load integer TIFF into ArcPro, use tool 'Build Raster Attribute Table'
    - Once attribute table is created NDVI values are associated with "Value" field
- The estimate of total vegetation returned in GEE uses an NDVI threshold of 0.2 (scaled to 200). With the differences between NAIP and Sentinel-2 imagery, the higher NDVI threshold seemed appropriate
- For May 2022 and these parameters, the total vegetation estimate is 9,575 acres

Reproducibility

The workflows described above are utilized in Google Earth Engine (GEE), a cloud-based platform that includes an enormous catalog of publicly available satellite imagery and built-in tools for geospatial data analysis. It is free to use for research, education, or by nonprofits. Registration is required. The script includes code that exports tree canopy area and vegetation area as a shapefile and a GeoTIFF raster that can be uploaded into a GIS project.

The scripts can be re-run in GEE with new data or with different NDVI parameters to create new estimates.

Uncertainty and Error Sources

This estimate provides a robust, repeatable method for tracking vegetation and tree canopy change in San Francisco using publicly available remote sensing data. While the methods are grounded in established workflows and refined for the 2022/2023 data, users should interpret the results as approximations rather than precise measurements. Given the known sources of uncertainty—such as timing gaps between data sources, variation in imagery quality, and the challenges of mapping in urban environments—the estimate is most valuable for identifying trends over time, prioritizing areas for further field verification, or guiding planning decisions at a citywide or neighborhood scale. Some sources of uncertainty are outlined below.

NAIP imagery has some of the best available spatial resolution of imagery datasets needed for discerning vegetation in an urban environment. However, it still aggregates real world conditions into 1 meter square pixels. Mixed pixels that include both vegetation and non-vegetation features are either counted as one or the other.

NAIP imagery is a mosaic of different flyovers introducing variation. Image stitching artifacts can lower reflectance values.

NAIP imagery collected in May 2022, may miss peak greenness for certain vegetation types

Clipping NAIP imagery and vectorizing the image raster can also be a source of errors affecting the overall estimate.

Since tree canopy is classified as a height above 8 feet and an NDVI value of above 0.05, any vegetation at a height above ground beyond 8 feet will be classified as a tree. In San Francisco, this includes any rooftop/balcony plantings, and all of Salesforce Park.

Any trees that are below 8 feet in height before LiDAR collection, such as recently pruned trees, or trees that are not sufficiently leafed out at the time of NAIP imagery collection may have been missed.

Ideally LiDAR and NAIP collection occur at the same time, however there is an 11 month gap between when LiDAR data was collected (April 2023) and NAIP data was collected (May 2022) affecting the tree canopy estimate.